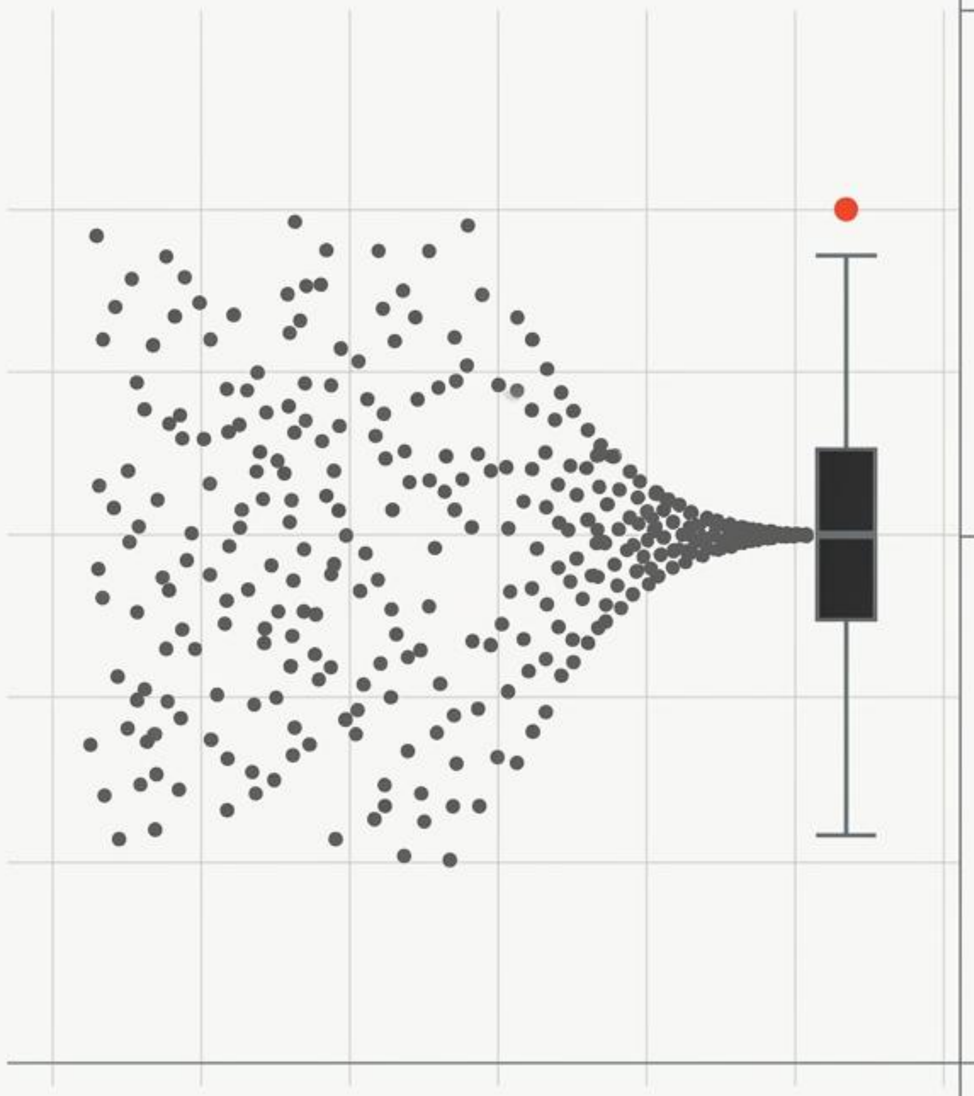
Descriptive Statistics: The Diagnostic Flow

Transforming raw datasets into structured visual heuristics.



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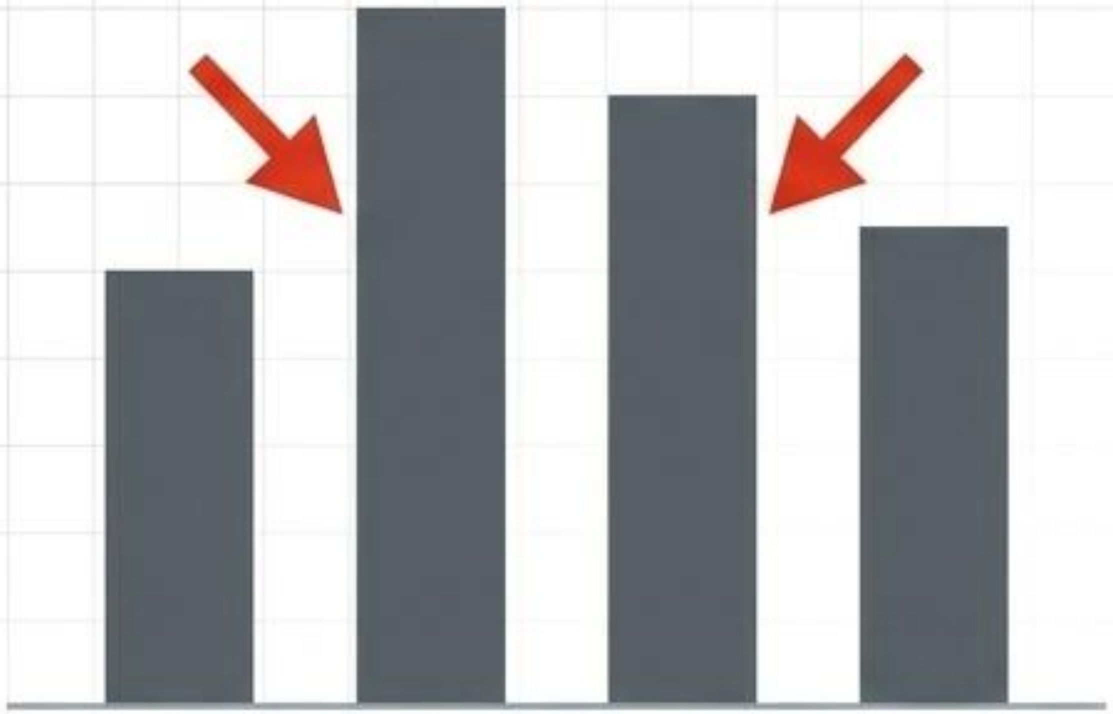
Step 1	Step 2	Step 3	Step 4	Step 5
The Foundation	**The Big Picture**	**The Anchor**	**The Footprint**	**The Exceptions**
(Classification)	(Visualization)	(Central Tendency)	(Spread)	(Outliers)
What kind of data are we looking at?	What is its shape?	Where is the middle?	How far does it reach?	What do we do with anomalies?

You cannot visualize data until you classify it. You cannot choose a measure of spread until you know your central tendency.

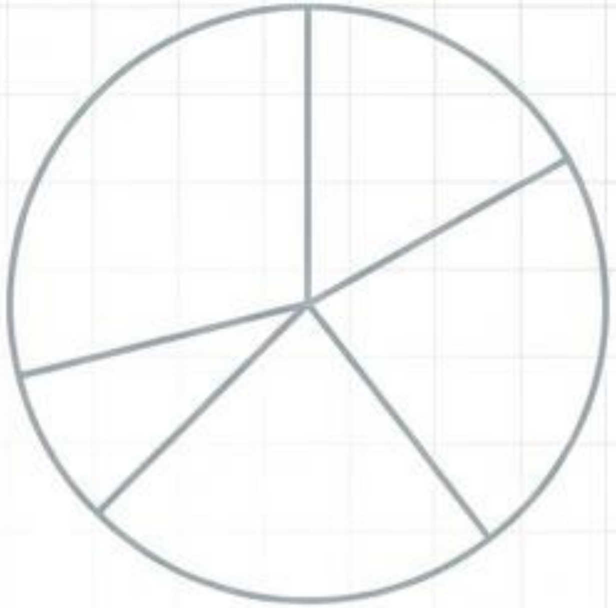
Categorical (Qualitative) Variable	Numerical Variable
Nominal Unordered names (e.g., Blood Group). No natural order; list by preference or size. Location measure: Mode.	**Continuous** Can take any value within a defined range (e.g., weight, blood pressure, temperature).
Ordinal Ordered stages (e.g., Disease Severity, Stage I to IV). **Crucial Rule:** Must maintain natural order in all tables and plots. Location measure: Mode or occasionally Median.	**Discrete** Distinct integer values determined by counting (e.g., number of hospital visits, number of cells).

Before calculating anything, create an ordered array (sorting raw data from smallest to largest) to quickly identify extremes and perform error checking.



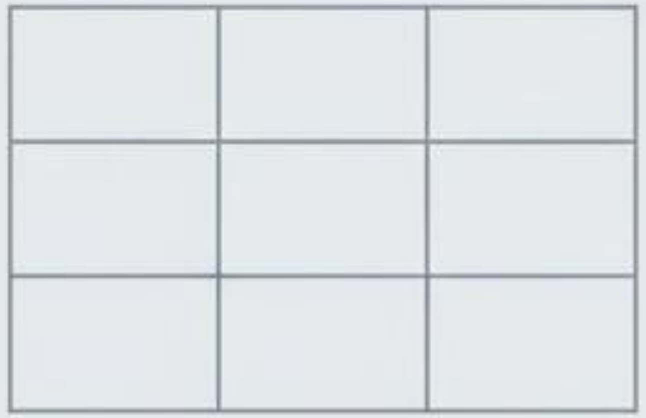
Bar Charts

Gaps must be left between bars to indicate discrete categories.



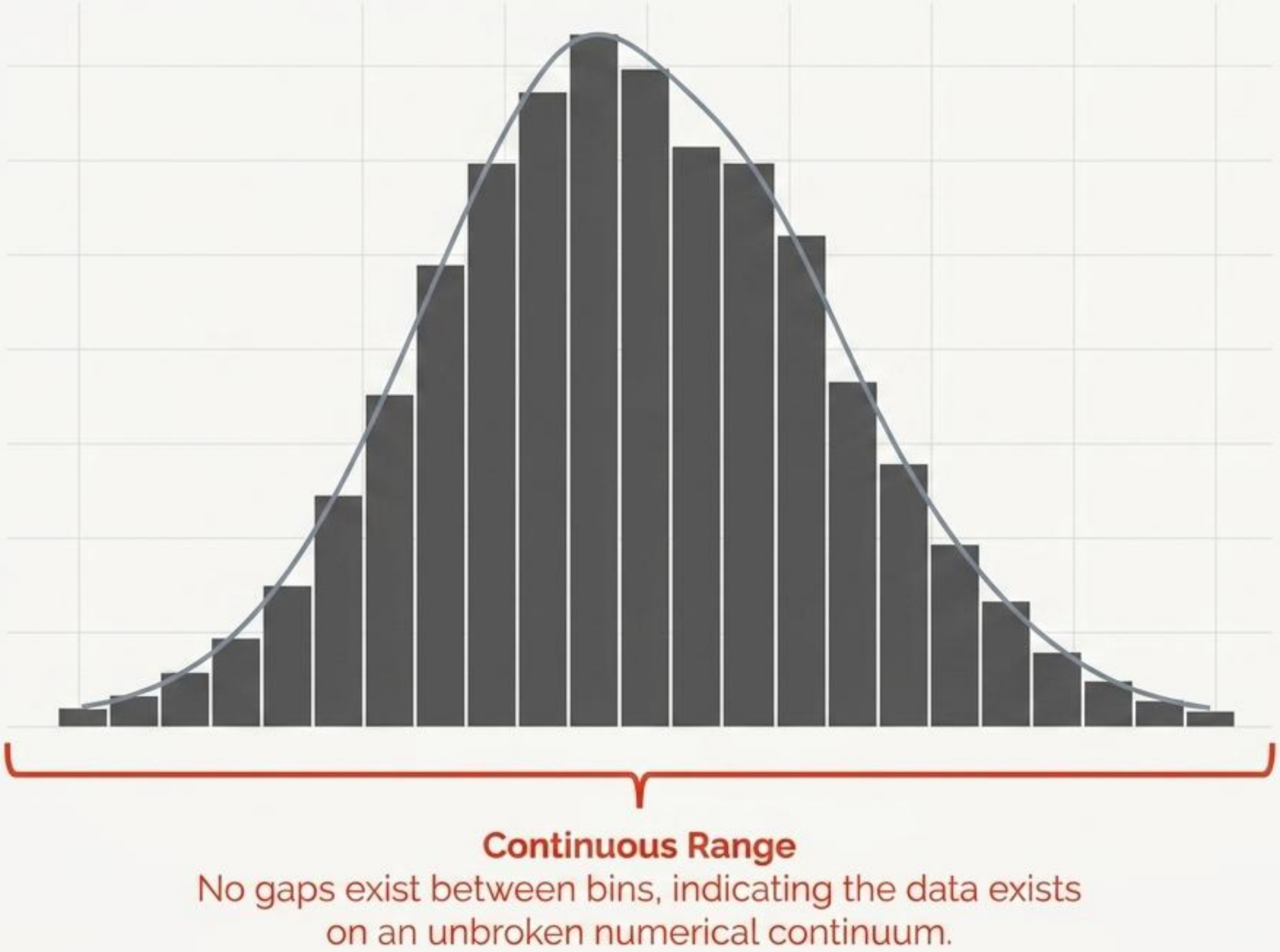
Pie Charts

Splits circles proportional to frequency. **Warning:** Often avoided when comparing groups, as the human eye struggles to compare angles relative to bar heights.



Summarizing Two Variables: Cross-Tabulations

To summarize two variables, place the exposure (independent) in the rows and the outcome (dependent) in the columns. Always report actual frequencies (*n*) alongside relative frequencies (percentages).



Visualizing Numerical Data

1. Use histograms to assess the overall shape, skewness, and symmetry of the continuous distribution.

2. Use scatter plots when exploring the mathematical association or correlation between two distinct numerical variables.

The Calculus of Centrality

Arithmetic Mean ($\bar{x}$)

Calculus: Sum of x_i / n

Uses all observations. Property: The sum of deviations from the mean always equals zero.

Median

The middle value of an ordered set (or average of two middle values if n is even).

Highly robust to extreme outliers.

Mode

Determined by counting. The most frequent category. Rarely used for continuous numerical variables.

Geometric Mean

Calculus: antilog($\bar{u}$) where $\bar{u}$ is the mean of log values.

Used specifically for positively skewed biological measurements (e.g., triglyceride levels).

Weighted Mean

Calculus: Sum of (w_i * x_i) / Sum of w_i

Used when observations have varying degrees of importance or differing sample sizes.

Trimmed Mean

Discards a fixed percentage (10-20%) from the extreme top and bottom before averaging, offering a robust compromise.

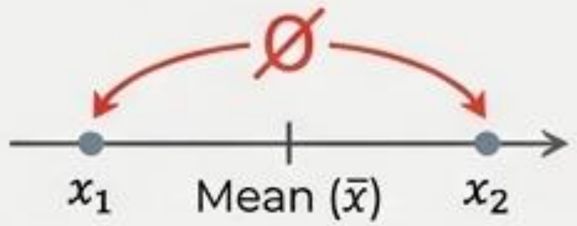
Arithmetic Mean	Median
Sensitivity: Distorted by outliers and pulled toward long tails in skewed data.	**Sensitivity:** Robust. Unaffected by extreme values.
Data Usage: Uses all data values.	**Data Usage:** Ignores most of the information in the dataset.
Mathematical Utility: Essential for inferential procedures (t-tests, confidence intervals) and deriving total budgets/sums.	**Mathematical Utility:** Not algebraically defined; complicated sampling distribution.
Context: Symmetrical, Normal distributions (e.g., healthy blood pressure).	**Context:** Income, survival times, disease rates, or severely non-Normal data.

Synthesis Tip: If data is skewed or you are unsure, quote both the Mean and the Median to provide a complete picture of location.

The Calculus of Spread: From Variance to Standard Deviation

Step 1: The Problem of Deviations

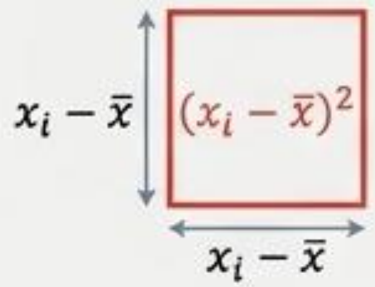
You cannot average raw deviations from the mean—positive and negative differences cancel exactly to zero.



Step 2: The Variance (s^2)

Formula: **s^2 = Sum of $(x\_i - \bar{x})^2$ / $(n - 1)$**

Annotation: We square the deviations to make them positive. We divide by n-1 (degrees of freedom) to accurately estimate the population variance.



Step 3: The Standard Deviation (s or SD)

Formula: **$s = \sqrt{s^2}$**

Annotation: We take the square root to return the spread to the original units of the data (e.g., kg instead of kg^2).

The Normal Rule: For bell-shaped Normal data, ~95% of observations lie within $\bar{x} \pm 2SD$.

Percentile-Based Spread
For Skewed Data

Concept: The Interquartile Range (IQR)

Calculus: IQR = Q3 - Q1

Application: Represents the central 50% of the data. Highly **robust** to extreme values.

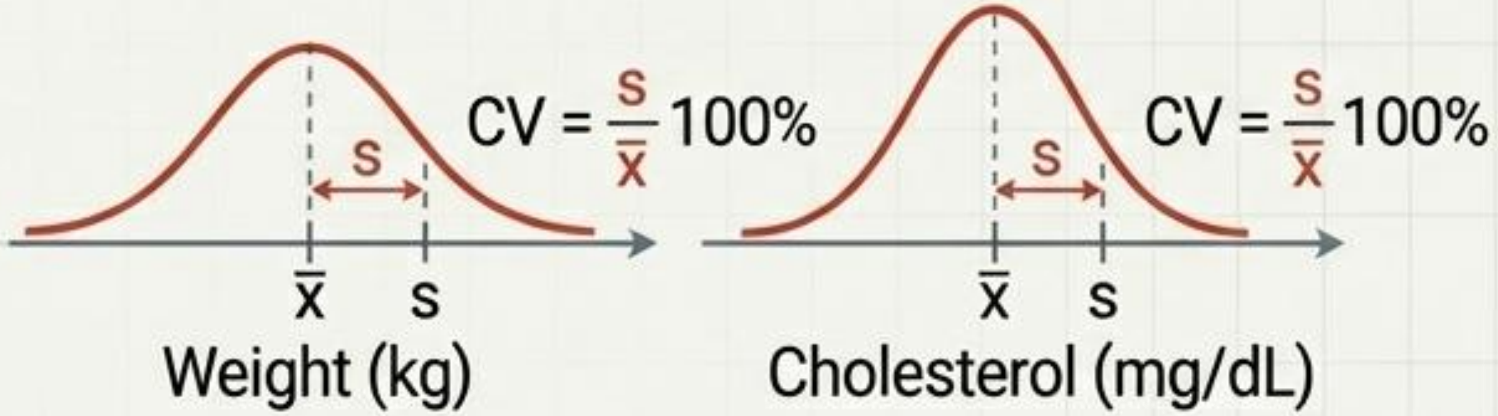
Note: The Interdecile Range (10th to 90th percentile) describes the **central 80%**.

Relative Dispersion
Comparing Scales

Concept: Coefficient of Variation (CV)

Calculus: CV = (s / $\bar{x}$) * 100%

Application: A dimensionless measure. Expresses standard deviation as a **percentage of the mean**. Used to determine if weight (kg) is **relatively more variable** than cholesterol (mg/dL) in the same population.



NotebookLM

Tukey's Five-Number Summary

3. Median (Q2)
(Divides dataset equally)

2. First Quartile (Q1)
(Lower Hinge: Median of the lower half)

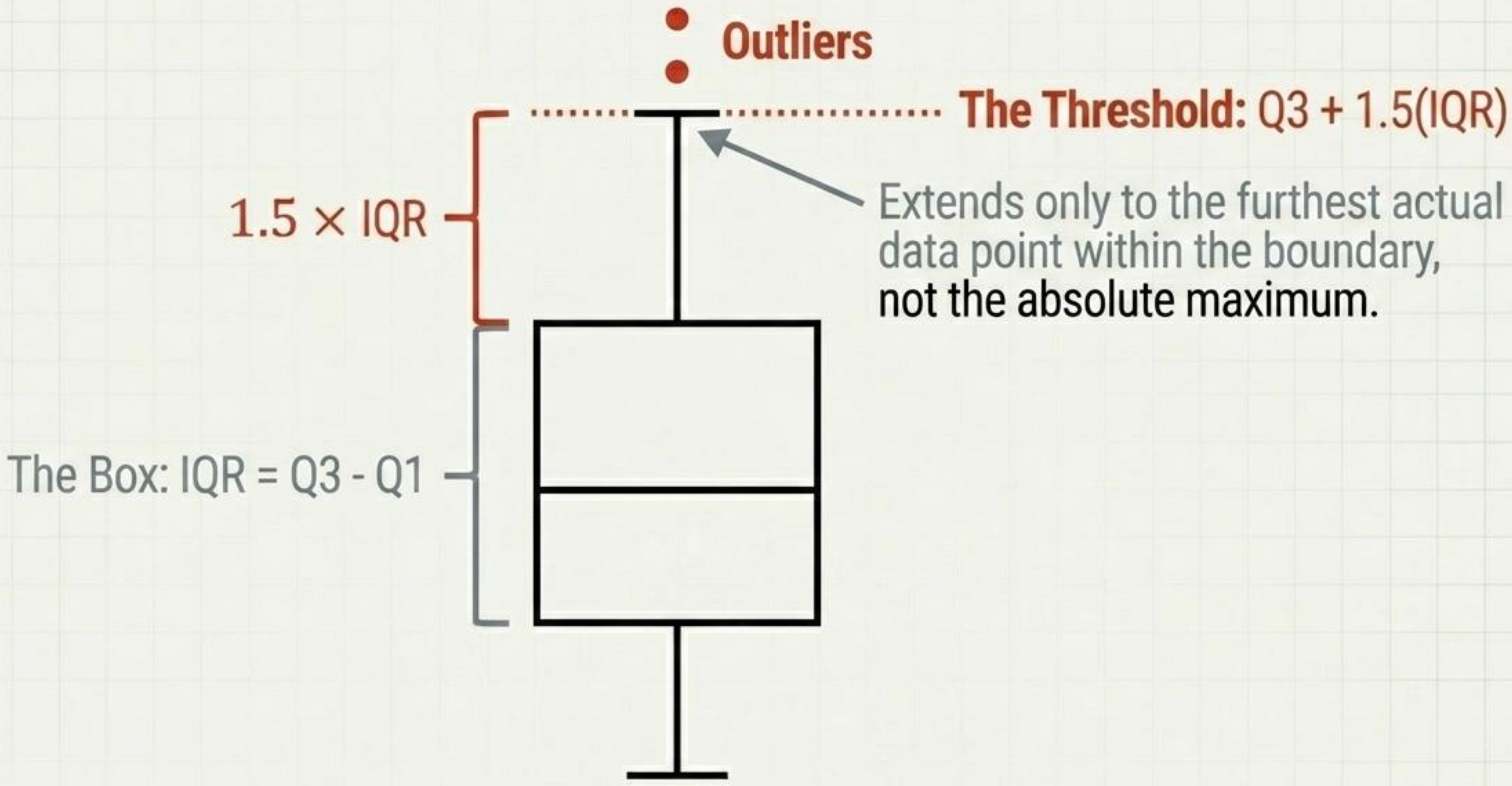
4. Third Quartile (Q3)
(Upper Hinge: Median of the upper half)

1. Minimum
(Smallest value)

5. Maximum
(Largest value)

Key Takeaway: Designed for **exploratory data analysis (EDA)**. Because it relies on mediais and hinges, it is inherently **robust to distortion**, making it perfect for **non-Normal data** and initial screening.

Boxplot



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Handling Identified Outliers

(Never automatically discard)

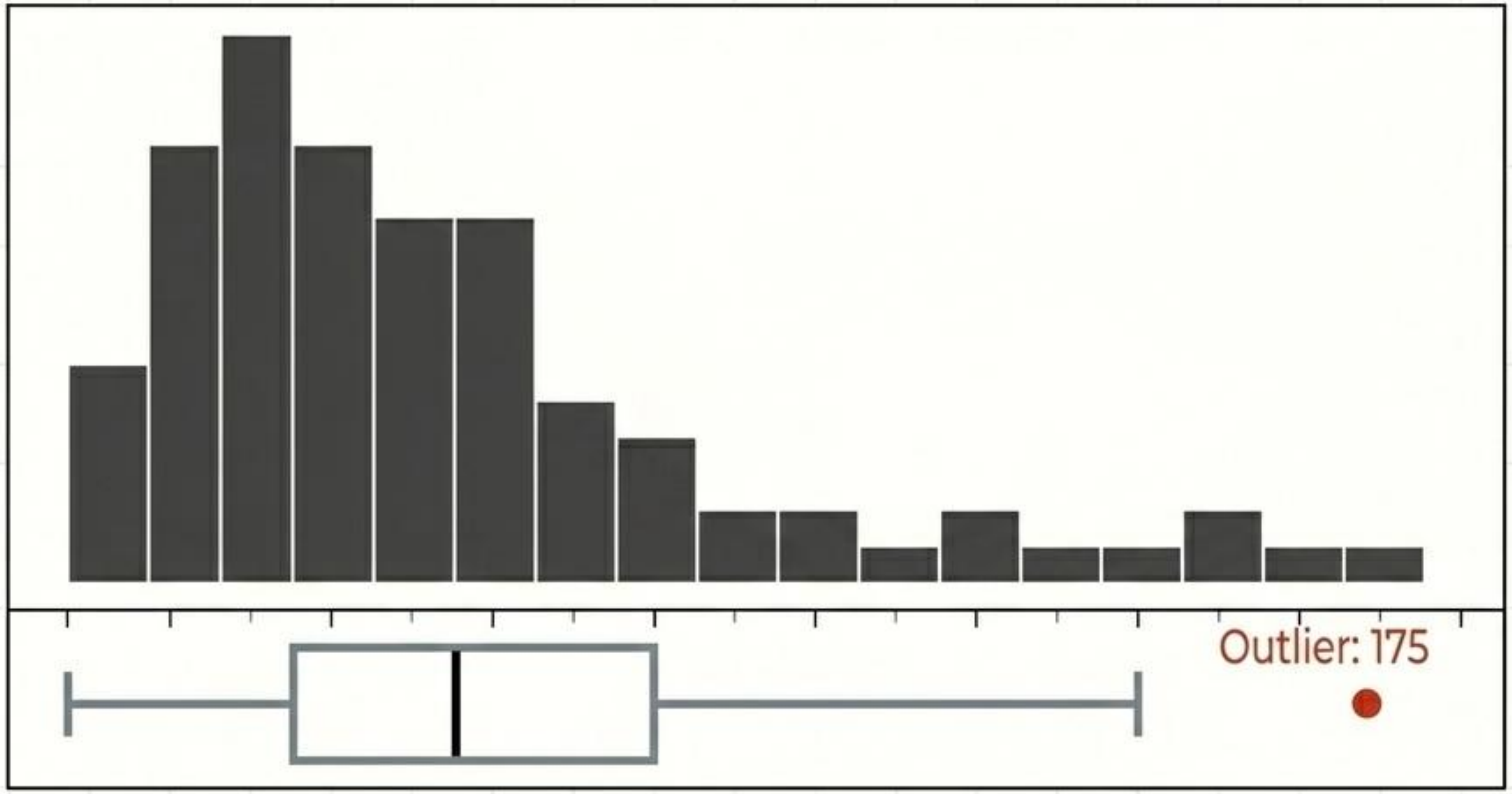
1. Check for Errors
Is it a data entry mistake? Check for misplaced decimal points or incorrect unit measurements.

2. Investigate Genuine Extremes
Is the value correct? It may represent a real physiological extreme (e.g., an exceptionally tall person) that contains vital population information.

3. Perform Sensitivity Analysis
Does it change the conclusion? Run the statistical test twice—both with and without the outlier—to observe its influence on the final analytical outcome.

The Diagnostic Flow Applied

Context: Systolic Blood Pressure (SBP) in 20 Patients. (**Data Type:** Continuous Numerical).



Centrality Focus

The Mean is **131.3**. However, pulled right by the outlier, the Median of **130** is the more typical '**Anchor**'.

Dispersion Focus

The Standard Deviation is **16.3**. The robust **IQR** (describing the middle 50%) is **17.5**.

Key Insight: A single visual dashboard instantly communicates the shape, location, spread, and exceptions of the raw data, proving the power of descriptive statistics.